

NovaShyld Cybersecurity Internship

Task 2: Lab Setup & Security Environment Configuration

Prepared By

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Lab Environment

Kali Linux • Metasploitable2 • DVWA • Docker • VirtualBox

August 2026

https://github.com/Reddy-hub-com/NovaShyld_Task_2

https://github.com/Reddy-hub-com/NovaShyld_Task_2.git

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1. Introduction

This task focuses on establishing a controlled cybersecurity laboratory environment for vulnerability assessment and penetration-testing activities. Kali Linux is used as the attacker/test system, while Metasploitable2 and DVWA provide deliberately vulnerable targets/applications. VirtualBox provides the virtualized environment and Docker is used to run DVWA.

2. Objectives

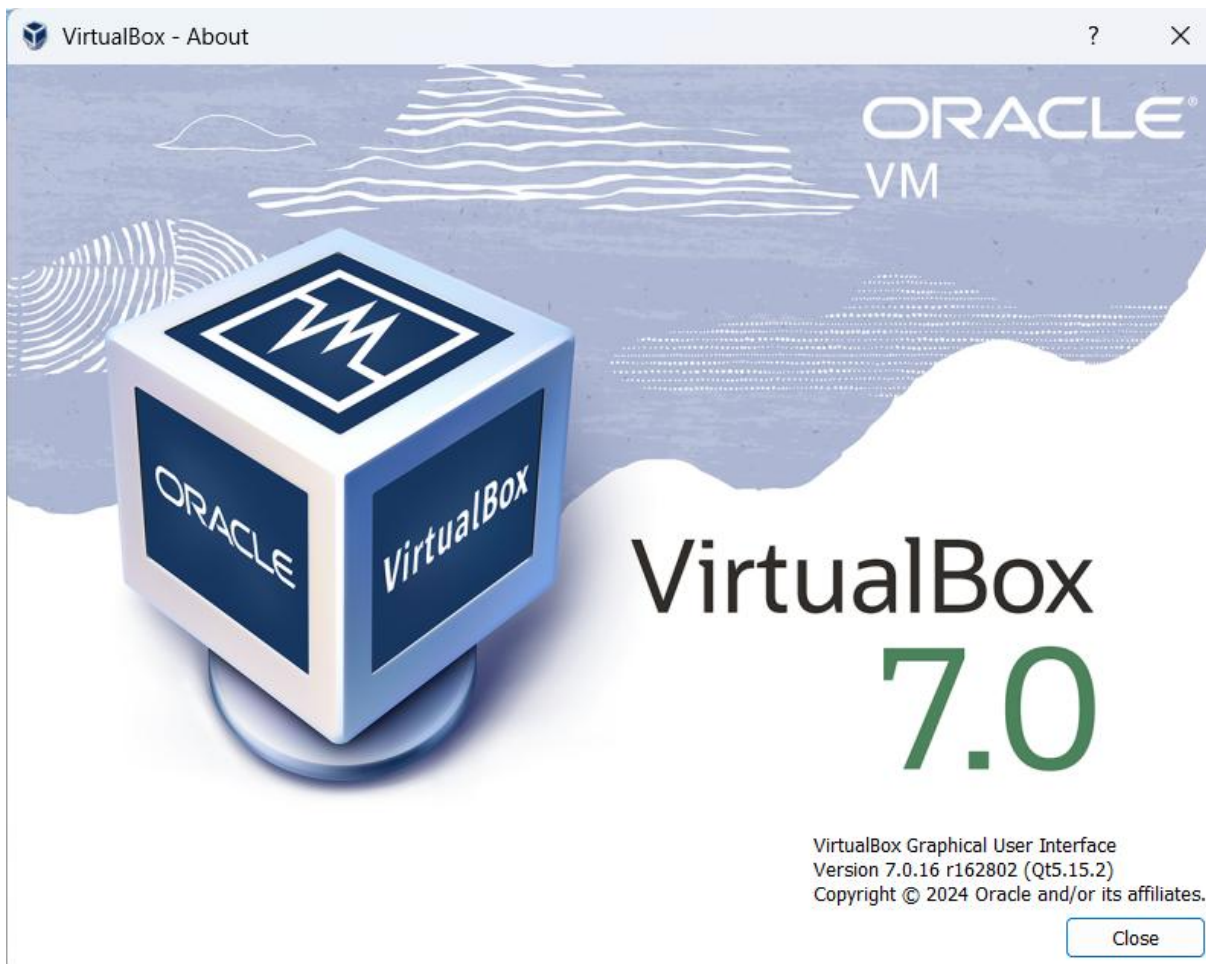
- Install and configure Kali Linux as the attacker environment.
- Deploy a vulnerable target environment using Metasploitable2.
- Deploy DVWA as a vulnerable web application using Docker.
- Configure communication between the attacker and target systems.
- Verify attacker–target connectivity.
- Document the laboratory configuration with screenshots.
- Prepare a network configuration diagram.

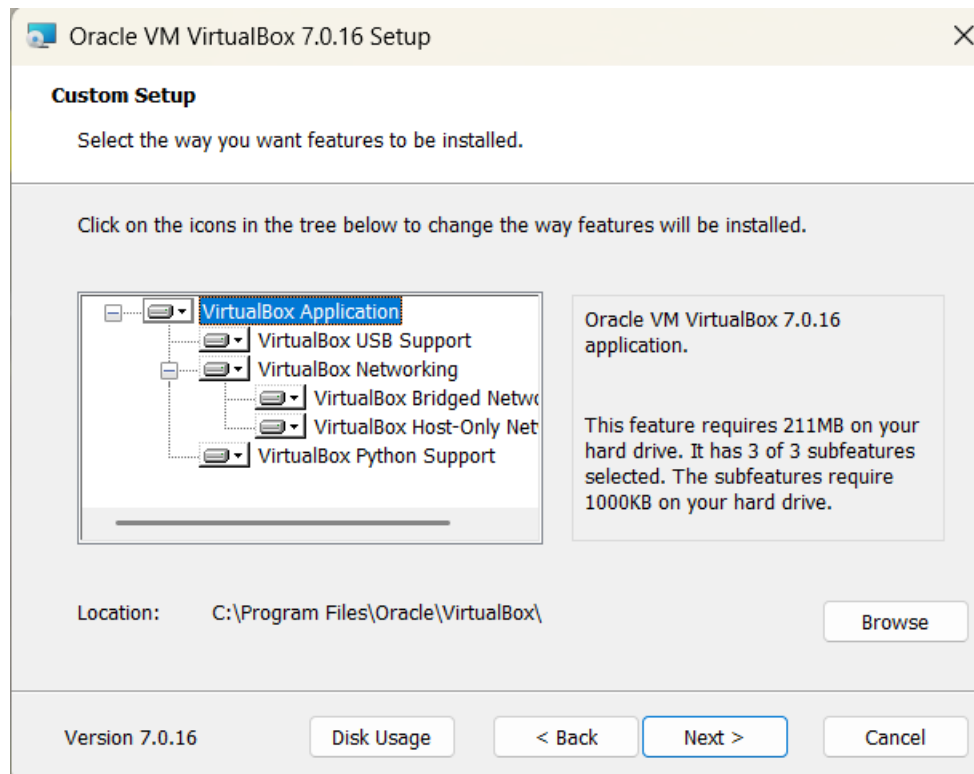
3. Lab Environment

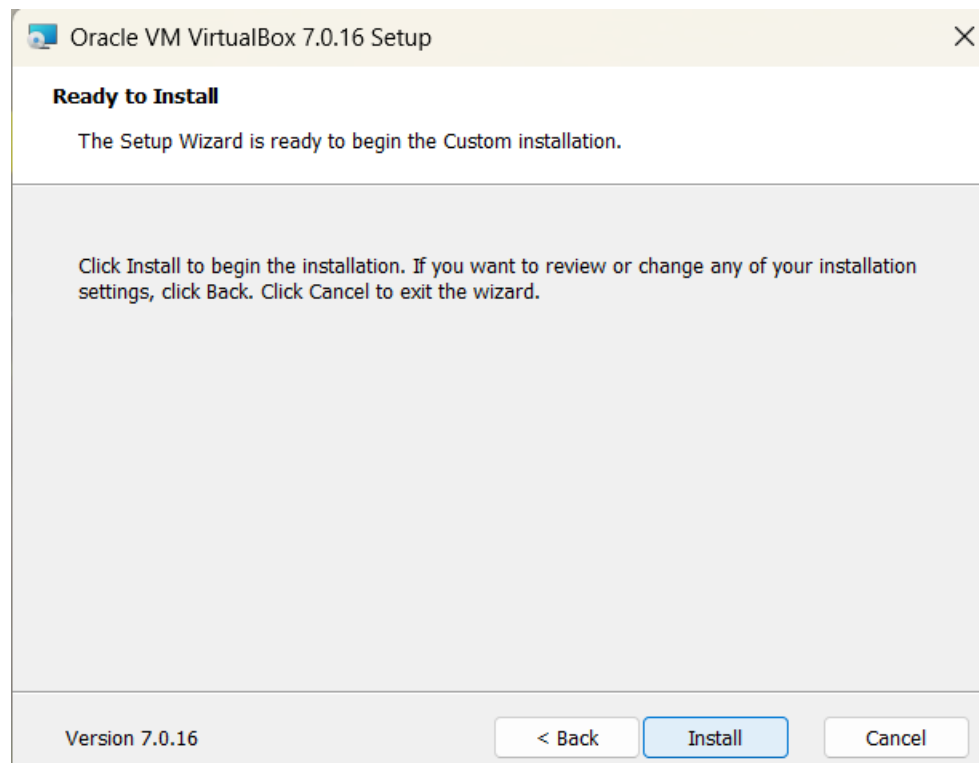
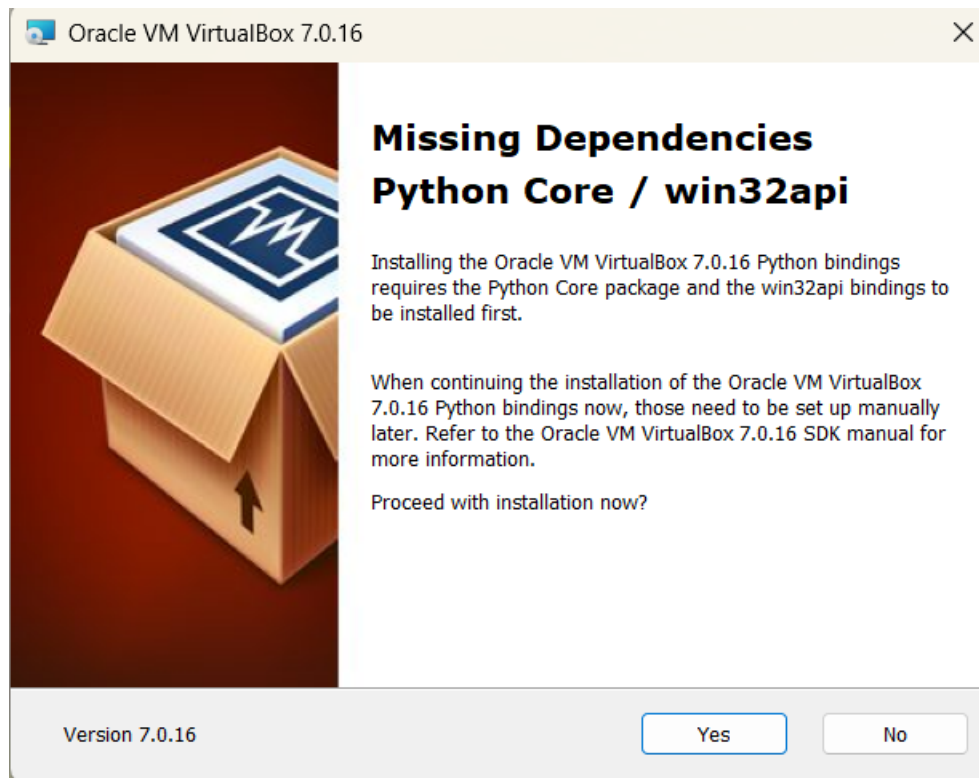
Component	Role	Status
VirtualBox	Virtualization platform	Configured
Kali Linux	Attacker / security testing system	Configured
Metasploitable2	Vulnerable target	Running / reachable
Docker	Container platform	Running
DVWA	Vulnerable web application	Accessible
Firefox	Web browser for DVWA	Verified

4. Activity 1 – Kali Linux Installation and Configuration

Kali Linux was configured as the primary attacker/test machine for the cybersecurity laboratory. The virtual machine was prepared using VirtualBox and used for network verification and subsequent security testing activities.







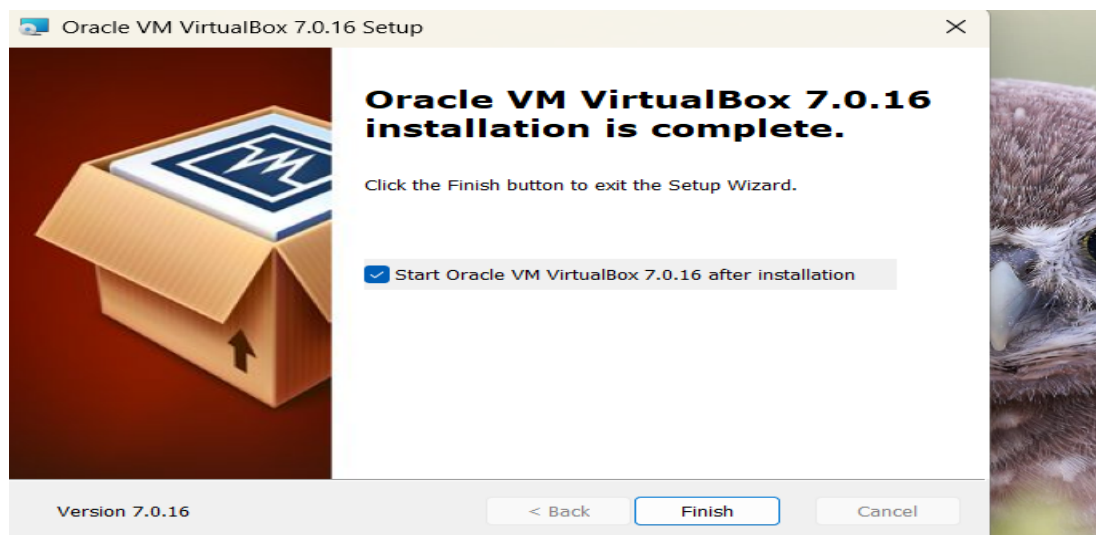
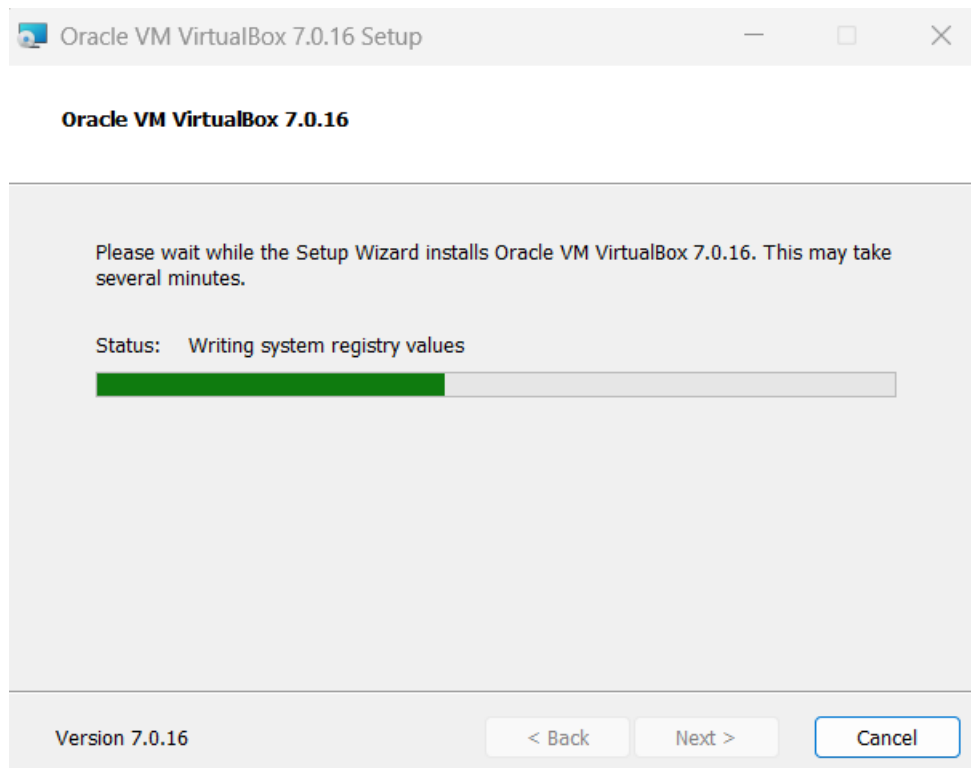
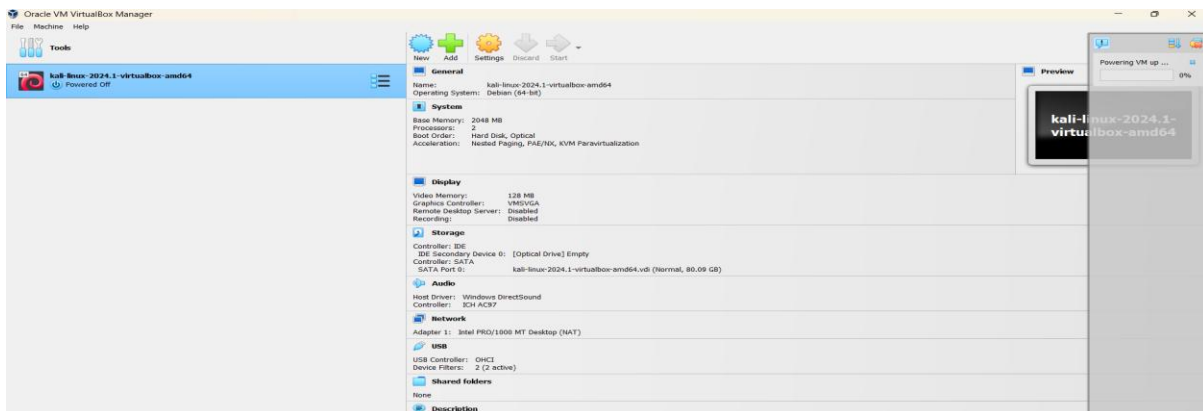
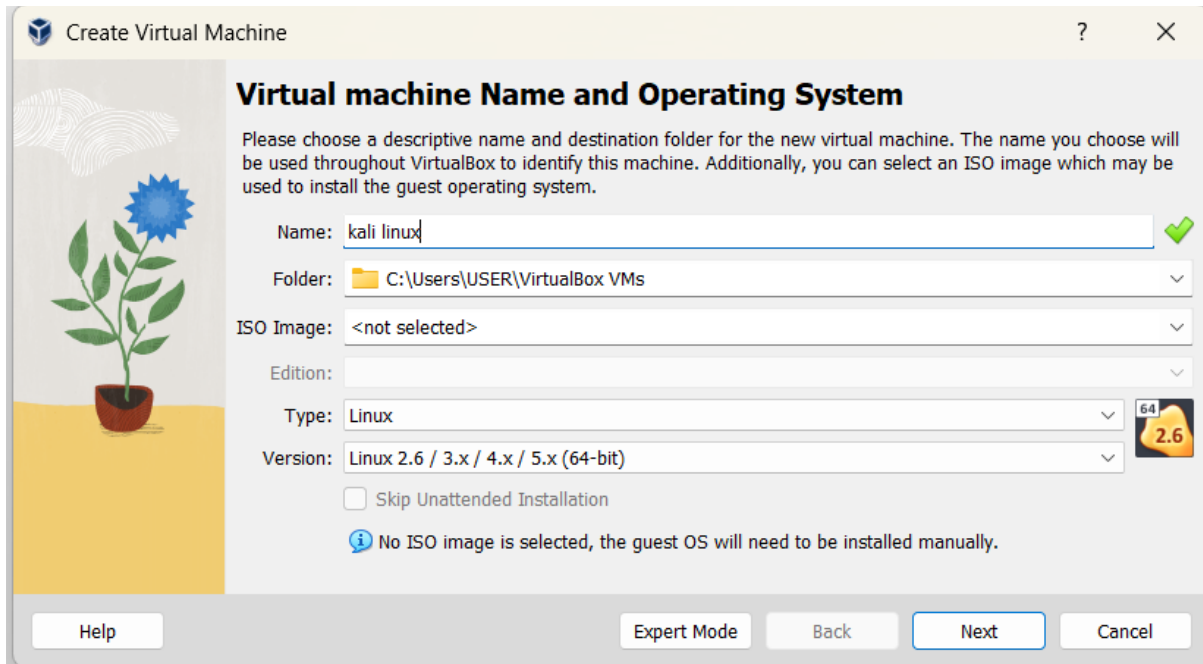
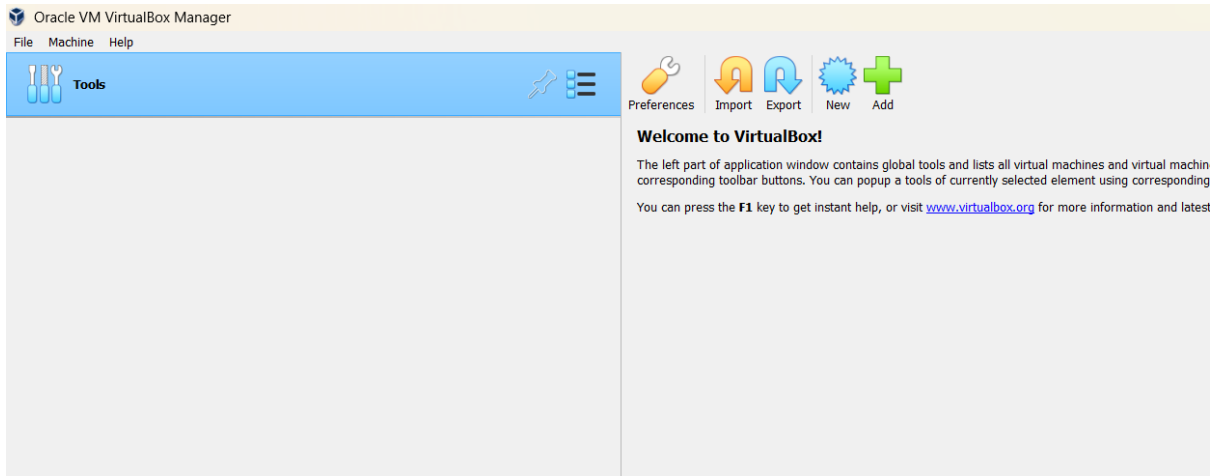
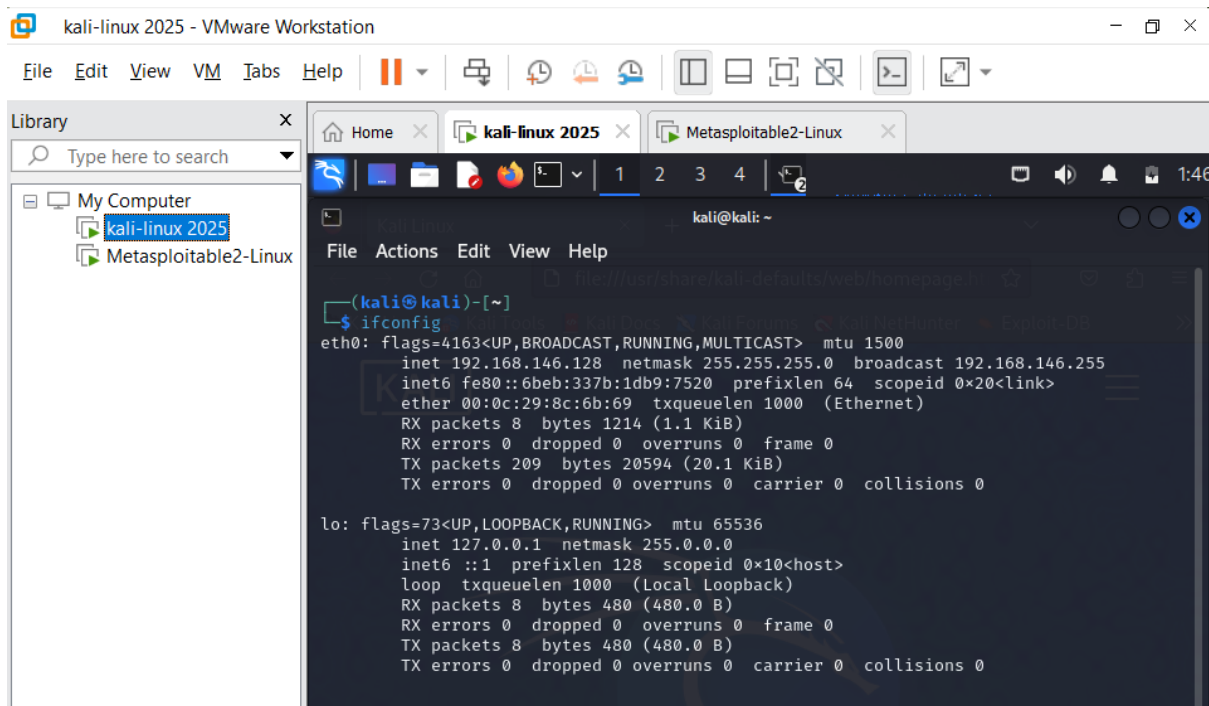
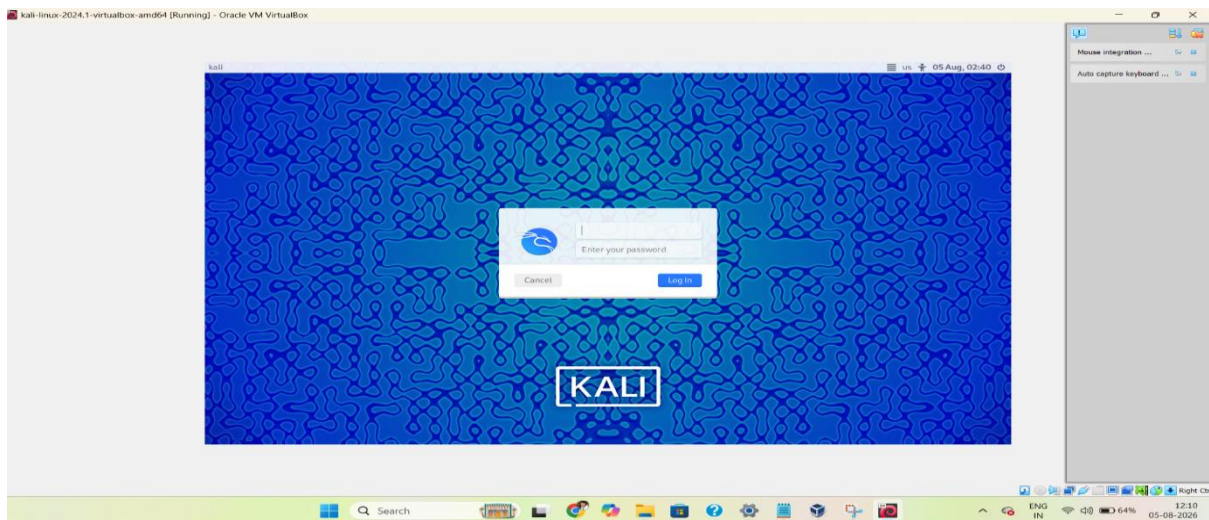


Figure 1: Kali Linux virtual machine configured as the attacker/test environment.





5. Activity 2 – Vulnerable Target Deployment

5.1 Metasploitable2

Metasploitable2 was deployed as a deliberately vulnerable Linux target. Its network address was identified and connectivity from Kali Linux was subsequently verified.

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Metasploitable Files

Metasploitable is an intentionally vulnerable Linux virtual machine

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metasploitable-linux-2.0.0.zip (865.1 MB)

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Name	Date modified	Type	Size
Metasploitable.nvram	05-08-2026 19:52	NVRAM File	9 KB
Metasploitable	05-08-2026 19:52	Virtual Machine Di...	18,80,512 ...
Metasploitable.vmsd	05-08-2026 19:52	VMSD File	0 KB
Metasploitable.vmx	05-08-2026 19:52	VMX File	3 KB
Metasploitable.vmxs	05-08-2026 19:52	VMXF File	1 KB

Create Virtual Machine

?

×

Summary

The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.

Machine Name and OS Type

Machine Name

Machine Folder

ISO Image

Guest OS Type

Hardware

Base Memory

Processor(s)

EFI Enable

Disk

Attached Disk

Help

Back

Finish

Cancel

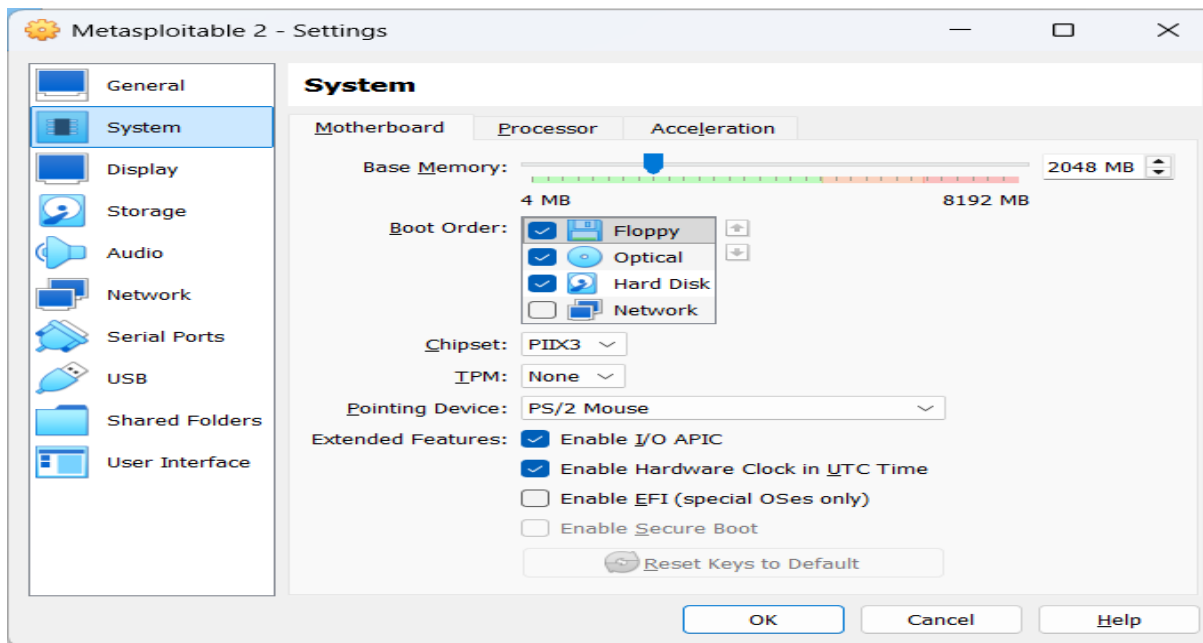
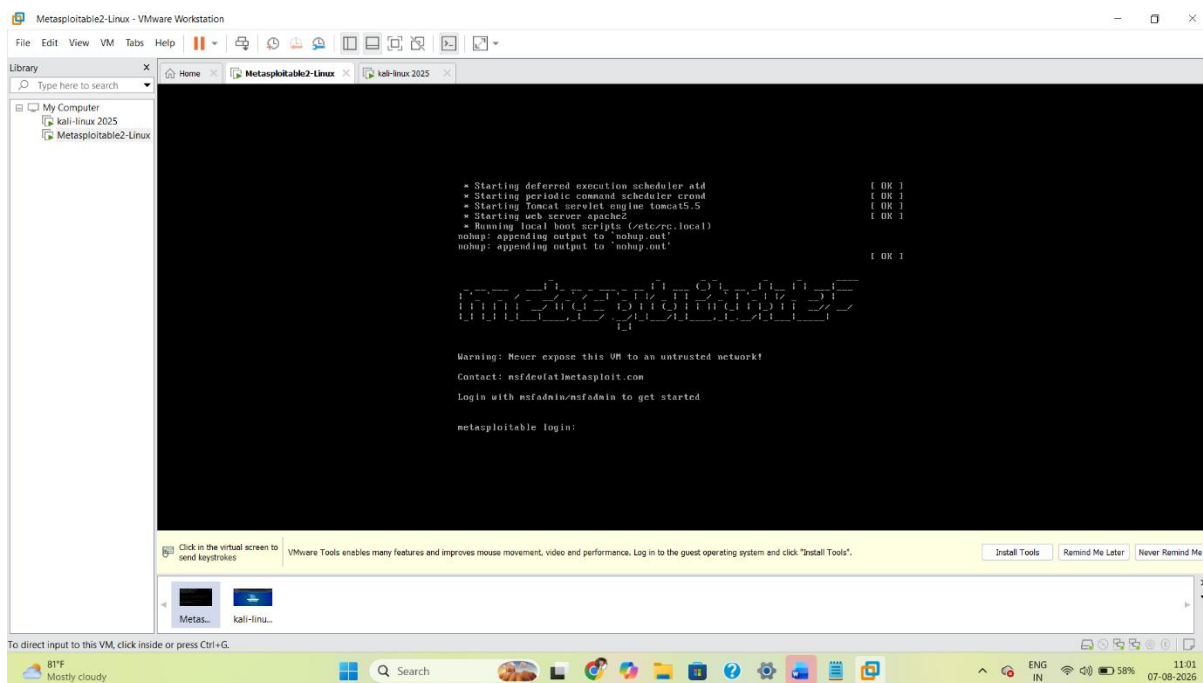


Figure 2: Metasploitable2 vulnerable target system.



The confirmed Metasploitable2 IP address is:

192.168.146.129

```

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:a5:52:9d
          inet addr:192.168.146.129  Bcast:192.168.146.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fea5:529d/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:13 errors:0 dropped:0 overruns:0 frame:0
          TX packets:48 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1616 (1.5 KB)  TX bytes:5352 (5.2 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:96 errors:0 dropped:0 overruns:0 frame:0
          TX packets:96 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21437 (20.9 KB)  TX bytes:21437 (20.9 KB)

```

Figure 3: Metasploitable2 IP address identification.

6. DVWA Deployment

6.1 Docker and DVWA

Docker was installed and configured in Kali Linux. The DVWA Docker image was pulled and the container was started. The running container was verified before accessing the web application.

DVWA image command used:

```
sudo docker pull vulnerables/web-dvwa
```

```

(kali@kali)-[~/Desktop]
└─$ docker --version
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed

(kali@kali)-[~/Desktop]
└─$
Check out what's new in the

(kali@kali)-[~/Desktop]
└─$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install
Executing: /usr/lib/systemd/systemd-sysv-install enable docker

```

Figure 4: Docker environment configured in Kali Linux.

```
(kali@kali):~/Desktop
└─$ docker --version
Docker version 20.10.7, build 00c70432

(kali@kali):~/Desktop
└─$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker

(kali@kali):~/Desktop
└─$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-08-07 03:21:25 EDT; 1min 56s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 12726 (dockerd)
      Tasks: 10
     Memory: 28.4M (peak: 30.6M)
        CPU: 731ms
     CGroup: /system.slice/docker.service
            └─12726 /usr/sbin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Aug 07 03:21:24 kali dockerd[12726]: time="2026-08-07T03:21:24.766821138-04:00" level=info msg="CDI directory does not exist, skipping: failed to monitor for changes: no such file or directory" dir=/var/run/cdi
Aug 07 03:21:24 kali dockerd[12726]: time="2026-08-07T03:21:24.876818446-04:00" level=info msg="Creating a containerd client" address=/run/containerd/containerd.sock timeout=1m0s
Aug 07 03:21:24 kali dockerd[12726]: time="2026-08-07T03:21:24.969930893-04:00" level=info msg="Loading containers: start."
Aug 07 03:21:25 kali dockerd[12726]: time="2026-08-07T03:21:25.749805510-04:00" level=info msg="Loading containers: done."
Aug 07 03:21:25 kali dockerd[12726]: time="2026-08-07T03:21:25.814368846-04:00" level=info msg="Docker daemon" commit=3ef3c87c883be49a2e8097bcfd62952fced7eb containerd-snapshotter=false storage-driver=overlayfs2
Aug 07 03:21:25 kali dockerd[12726]: time="2026-08-07T03:21:25.814549049-04:00" level=info msg="Initializing buildkit"
Aug 07 03:21:25 kali dockerd[12726]: time="2026-08-07T03:21:25.857219146-04:00" level=info msg="Completed buildkit initialization"
Aug 07 03:21:25 kali dockerd[12726]: time="2026-08-07T03:21:25.876268370-04:00" level=info msg="Daemon has completed initialization"
Aug 07 03:21:25 kali dockerd[12726]: time="2026-08-07T03:21:25.876421666-04:00" level=info msg="API listen on /run/docker.sock"
Aug 07 03:21:25 kali systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-32/32 (END)
```

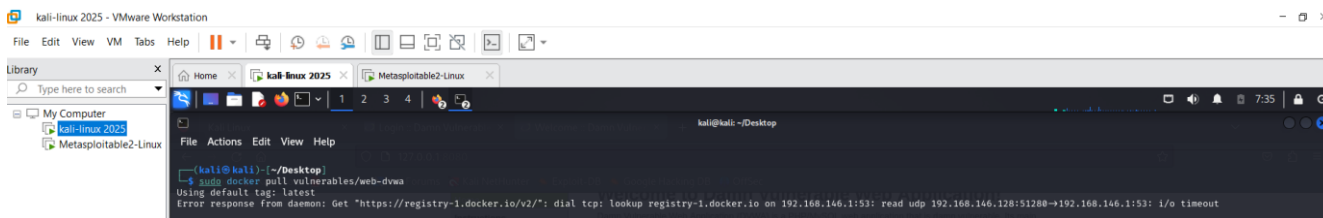


Figure 5: DVWA Docker image downloaded successfully.

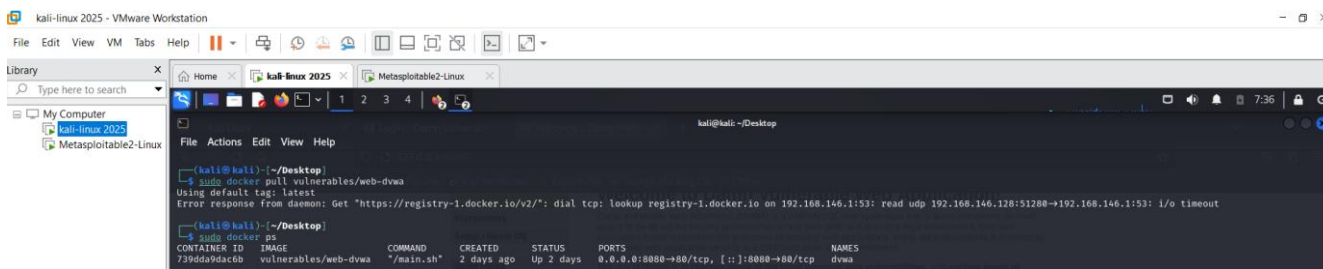


Figure 6: DVWA Docker container running successfully.

```
(kali@kali):~/Desktop
└─$ sudo usermod -s /usr/sbin/dmccs $USER

(kali@kali):~/Desktop
└─$ sudo docker pull vulnerables/web-dvwa
Using default tag: latest
latest: Pulling from vulnerables/web-dvwa
3e17deae6dc: Pull complete
0c57df616dbf: Pull complete
e805d18be401: Pull complete
e998ae5981d2: Pull complete
2c07db08257: Pull complete
6cfff35147f: Pull complete
090cfd63466: Pull complete
b30da332420: Pull complete
Digest: sha256:dae203f11646a86937bf04db0079ade295f426da68a92b4e3b181f337daa7
Status: Downloaded newer image for vulnerables/web-dvwa:latest
docker.io/vulnerables/web-dvwa:latest

(kali@kali):~/Desktop
└─$ sudo docker run -d --name dvwa -p 8080:80 vulnerables/web-dvwa
739dd9dac6b9b92abc19472807459909a6ecdcb81d9087ccf8169b7f91dbf0

(kali@kali):~/Desktop
└─$ sudo docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED              STATUS              PORTS              NAMES
739dd9dac6b   vulnerables/web-dvwa   "/main.sh"              40 seconds ago      Up 39 seconds      0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   dvwa

(kali@kali):~/Desktop
└─$
```

to direct input to this VM, move the mouse pointer inside or press Ctrl+G.

DVWA was accessed through:

<http://localhost:8080>

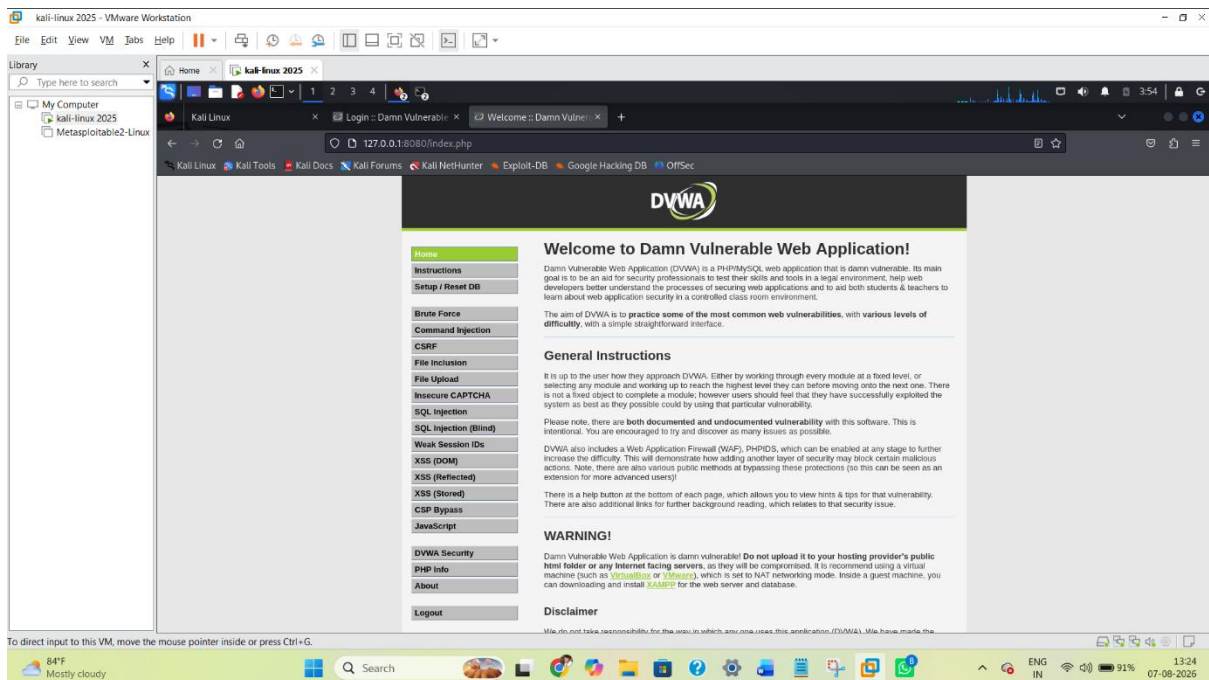
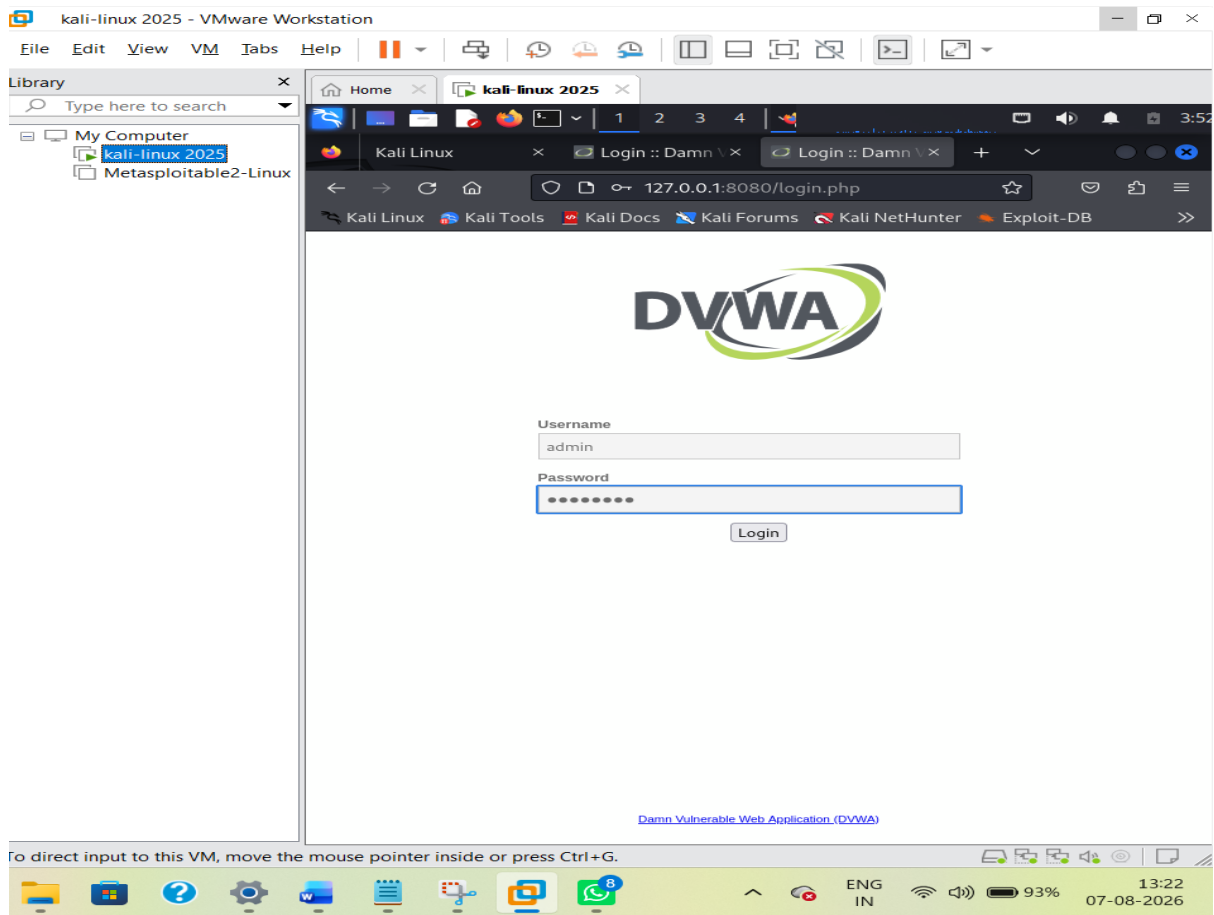
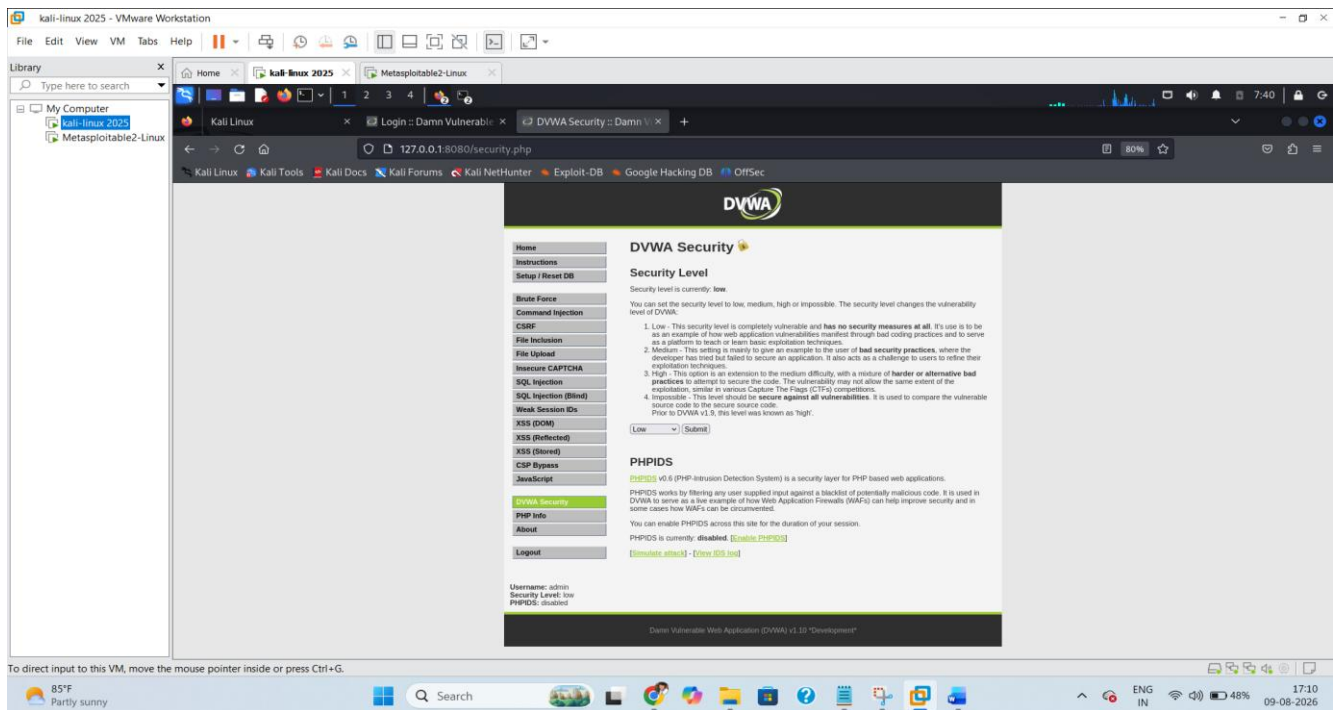


Figure 7: DVWA web application successfully accessible through Firefox.



The DVWA interface displayed the vulnerable application modules, including Brute Force, Command Injection, CSRF, File Inclusion, File Upload, SQL Injection, XSS (Reflected), XSS (Stored), and Weak Session IDs.

7. Activity 3 – Network Configuration

The virtual laboratory network was configured to allow the attacker and vulnerable target systems to communicate for authorized testing. The exact adapter/network name should be copied from the user's VirtualBox network settings rather than assumed.

System	Role	IP / Address	Network Details
Kali Linux	Attacker / Tester	192.168.146.128	[xxxxxxxxx]
Metasploitable2	Vulnerable Target	192.168.146.129	[xxxxxxxxx]
DVWA	Vulnerable Web App	localhost:8080 / Docker	Docker container

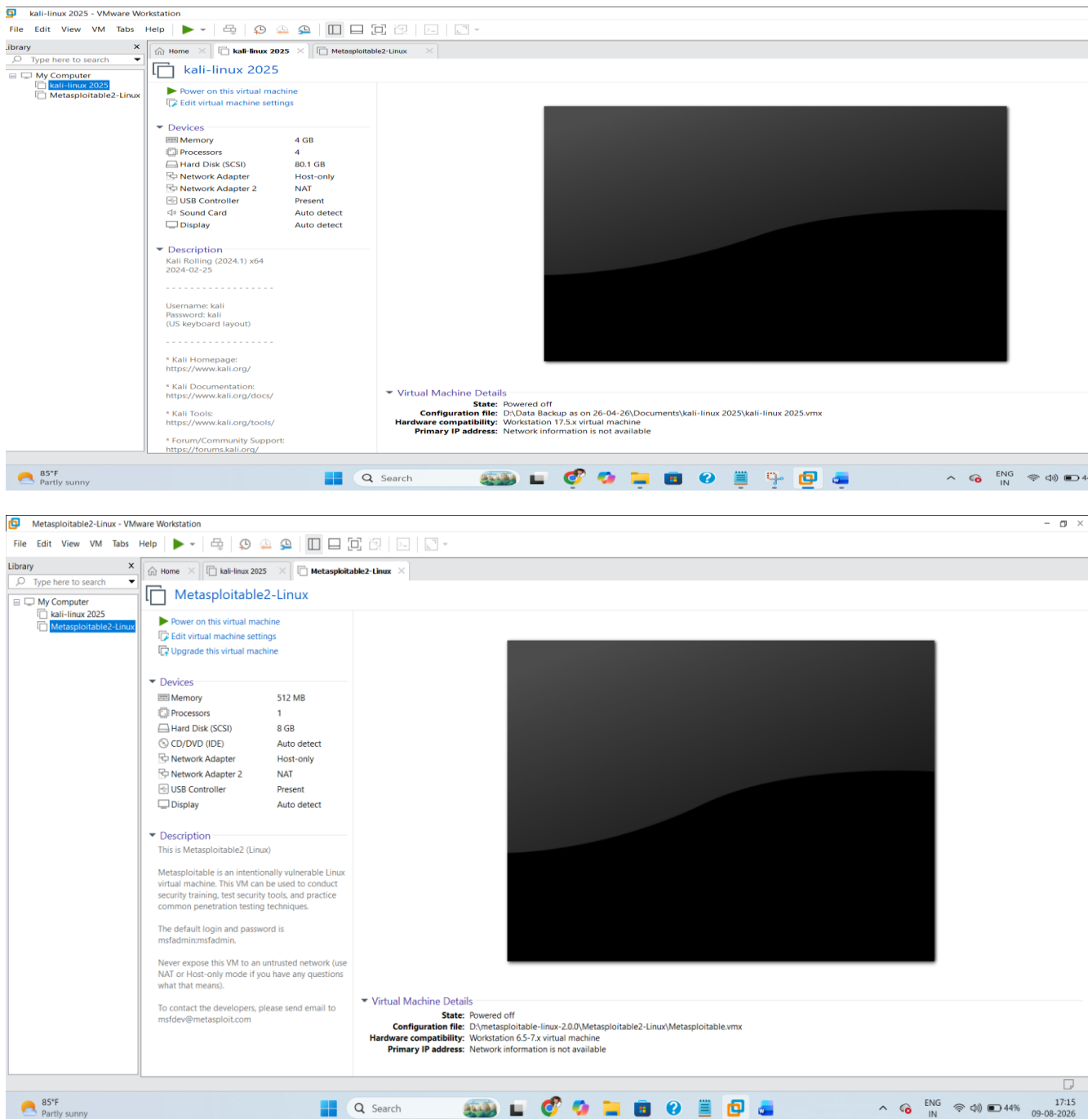


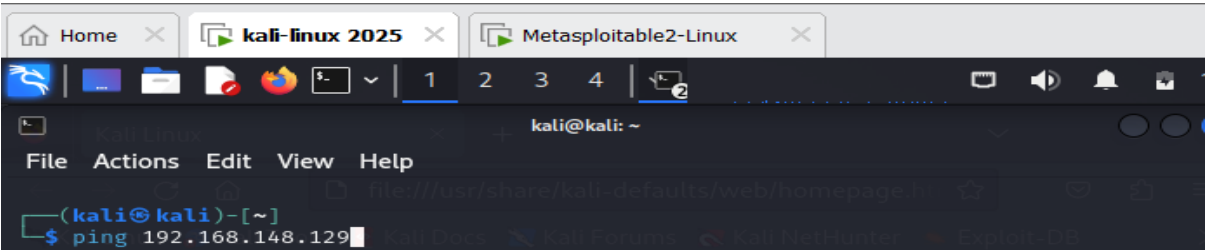
Figure 8: VirtualBox network configuration used for the laboratory.

8. Activity 4 – Attacker–Target Connectivity Verification

Connectivity between Kali Linux and Metasploitable2 was verified using ICMP ping. The test targeted the confirmed Metasploitable2 address 192.168.146.129.

```
ping -c 4 192.168.146.129
```

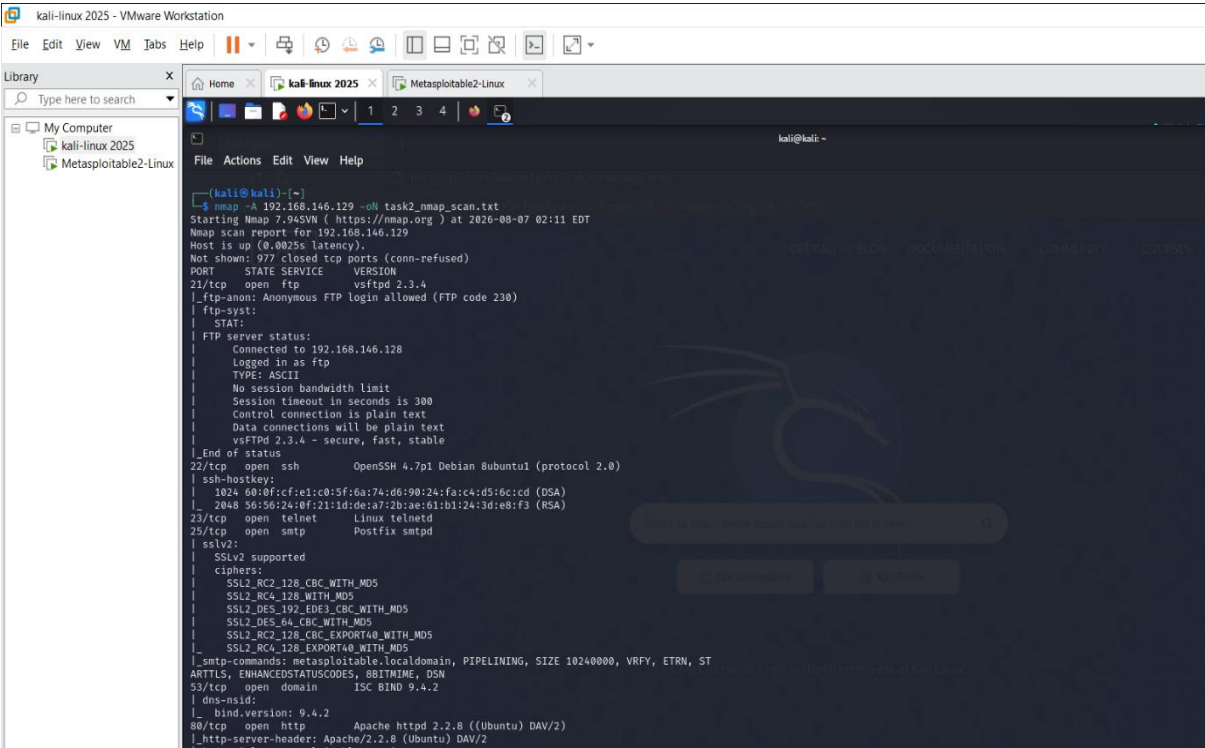

The test returned 4 packets received from 4 transmitted packets, with 0% packet loss. This confirms successful network connectivity between the Kali Linux attacker system and the Metasploitable2 target.



```
(kali@kali)-[~/Desktop]
$ ping -c 4 192.168.146.129
PING 192.168.146.129 (192.168.146.129) 56(84) bytes of data.
64 bytes from 192.168.146.129: icmp_seq=1 ttl=64 time=3.48 ms
64 bytes from 192.168.146.129: icmp_seq=2 ttl=64 time=0.958 ms
64 bytes from 192.168.146.129: icmp_seq=3 ttl=64 time=0.681 ms
64 bytes from 192.168.146.129: icmp_seq=4 ttl=64 time=0.528 ms

— 192.168.146.129 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3012ms
rtt min/avg/max/mdev = 0.528/1.412/3.483/1.205 ms
```

Figure 9: Successful Kali Linux to Metasploitable2 connectivity verification (0% packet loss)



9. Network Configuration Diagram

The following diagram documents the laboratory architecture. The Metasploitable2 address is the confirmed value from the connectivity test. The Kali IP and exact VirtualBox network name are intentionally left as fields to be filled from the user's captured configuration screenshots, avoiding unsupported assumptions.

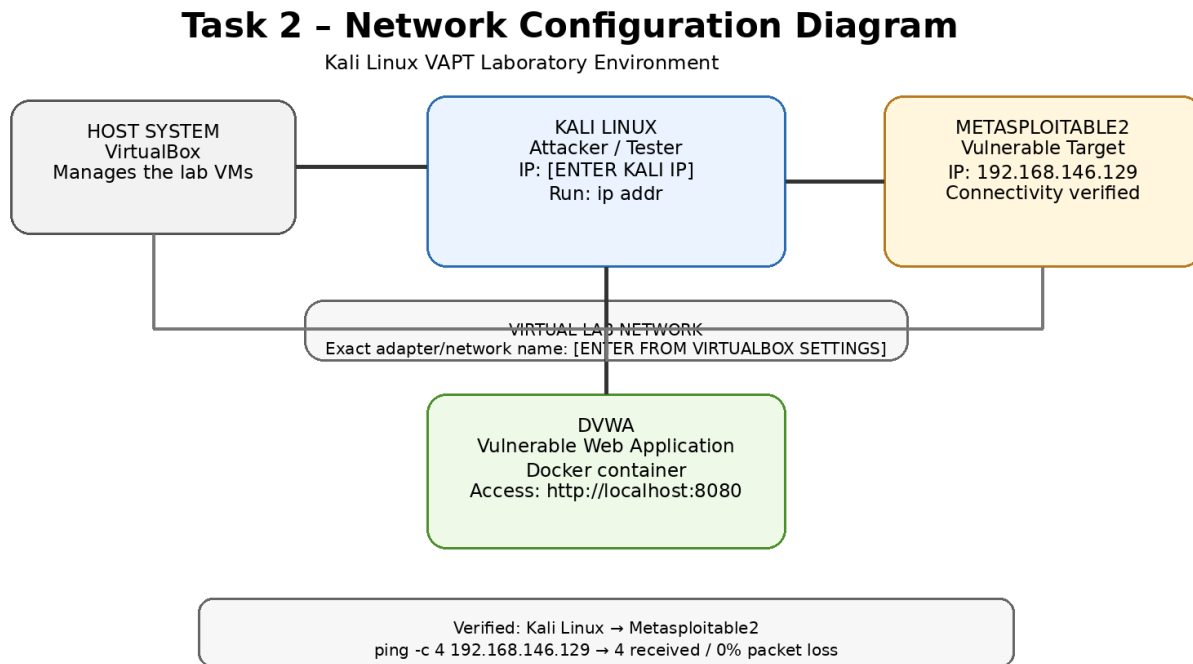


Figure 10: Task 2 network configuration diagram.

11. Results

- Kali Linux attacker environment was successfully configured.
- Metasploitable2 was successfully deployed as a vulnerable target.
- DVWA was successfully deployed using Docker.
- DVWA was accessible at <http://localhost:8080>.
- The expected DVWA vulnerable modules were visible.
- Metasploitable2 was identified at 192.168.146.129.
- Kali Linux successfully reached Metasploitable2.
- The connectivity test produced 4/4 replies and 0% packet loss.

12. Security Considerations

Metasploitable2 and DVWA are intentionally vulnerable systems and should be used only in an authorized, controlled laboratory environment. They should not be exposed directly to untrusted networks or the public Internet. Testing should remain within the configured internship lab.

13. Conclusion

Task 2 successfully established the core cybersecurity laboratory environment required for subsequent internship activities. Kali Linux was configured as the attacker/test system, Metasploitable2 was deployed as a vulnerable target, and DVWA was deployed using Docker as a vulnerable web application. DVWA was successfully accessed through the browser, and the Kali-to-Metasploitable2 connectivity test confirmed 4 packets received with 0% packet loss. The remaining submission work is to insert the user's already-captured setup screenshots and replace the clearly marked network placeholders with the exact values visible in those screenshots.